Gabriel Amoah Arthur

PhD Student, Statistics

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Professional Summary

Statistics PhD student with NSF/NIH-funded research in statistical genomics. Research interests span **statistical genomics**, **Bayesian methods for clinical trials**, and **high-dimensional data analysis**, motivated throughout by applying statistical methodology to problems in **public health and genetics**. Seeking a **2027 biostatistics / data science internship** where Bayesian modeling, survival analysis, and large-scale simulation translate into decisions about drug development or genomic diagnostics.

Education

Expected 2028 **Ph.D., Statistics**, *Iowa State University*, Ames, IA
Dissertation: *Probabilistic Consensus Sequence Inference of Viral Genomes via Profile Hidden Markov Models.*
Coursework: Advanced Statistical Inference, Measure-Theoretic Probability, Survival Analysis, Statistical Genetics, High-Dimensional Data Analysis, Statistical Computing.

2023–2026 **M.S.,Statistics**, *Iowa State University*, Ames,IA

2018–2022 **B.S., Statistics**, *Kwame Nkrumah University of Science and Technology*, Kumasi, Ghana

Work Experience

Jan 2026–Present **Graduate Research Assistant**, *Department of Statistics, Iowa State University*, Ames, IA
A Novel Profile HMM for Genotyping Viruses in Diagnostic Samples from Noisy Sequencing Reads
NSF/NIH-funded

- Designed a novel profile hidden Markov model that infers a *probabilistic* consensus sequence for viral genomes.
- Benchmarked the method head-to-head against the standard variant-calling stack (`iVar`, `Medaka`, `BCFtools`) across divergence categories, insertion/deletion scenarios, and sequencing-depth regimes.
- Established per-base posterior probability score (PBPS) as the primary evaluation metric, demonstrating improved accuracy in low-depth and high-divergence settings where conventional pipelines degrade.
- Built the full simulation and evaluation pipeline in R/Python; presented simulation findings to collaborators via LaTeX/Beamer technical seminars.

Aug 2023–Dec 2025 **Graduate Teaching Assistant**, *Iowa State University*, Ames, IA

- Served as **course instructor of record** for STAT 2260 (Business Statistics), delivering lectures and assessments to a full section of undergraduates.
- Led weekly computer lab sessions for STAT 101, guiding students through applied data analysis and statistical software.
- Graded written homework and provided individual support during office hours.

Nov 2022–Jul 2023 **Statistician (National Service)**, *Kumasi Center for Collaborative Research (KCCR), KNUST*, Kumasi, Ghana

- Analyzed **clinical trial data** supporting drug development studies.
- Managed case report form (CRF) data entry and quality control in **REDCap**, maintaining data integrity across study visits.

Oct 2021–Nov 2021 **Assistant Statistician (Intern)**, *Ghana Revenue Authority, Agona Swedru*, Ghana

- Performed transaction data entry and verified daily records against source documents to catch entry errors.
- Supported the lead statistician in compiling weekly and monthly reporting.

Aug 2017–Jul 2018 **Academic Tutor**, *Swedru Omega Preparatory School*, Agona Swedru, Ghana
Taught Mathematics and Religious & Moral Education to Primary 4–6 students.

Manuscripts in Preparation

In prep. **Arthur, G. A.**, **Dorman K.** et al. *A Novel Profile Hidden Markov Model for Genotyping Viruses in Diagnostic Samples from Noisy Sequencing Reads.* NSF/NIH-funded project targeted at *Oxford Bioinfoormatics Review*. *Manuscript in preparation.*

In prep. **Arthur, G. A.**, **Ayikambey D.** et al. *Variational Bayes Inference for the Log-Logistic Accelerated Failure Time Model with Interval-Censored Data.* Fast, deterministic alternative to MCMC for Bayesian survival analysis, with VB posteriors calibrated against MCMC and an accompanying R package; targeted at *Statistics and Computing*. *Manuscript in preparation.*

Research Projects

2026 **Fractional Stochastic Modified Equations & Optimal Control**, *Joint work with P. A. Tchoumo and K. Ogunsusi*

- Built a new mathematical model of the noise in stochastic gradient descent that allows the noise to carry memory over time, rather than assuming each step is independent.
- Used the model to derive adaptive learning-rate schedules and closed-form results on how memory changes the speed and stability of training.
- Developed a simulation and machine-learning pipeline in Python to estimate the memory parameter, validated by large-scale Monte Carlo study.
- Applied it to real data — gradient noise from a CIFAR-10 image classifier and financial time series — confirming measurable memory effects in both.

2026–Present **Variational Bayes Engine for Interim Monitoring in Time-to-Event Trials**, *Iowa State University*, Ames, IA

- Developing a variational Bayes inference engine for the log-logistic AFT model with **interval-censored** event times, extending existing right-censoring-only methods.
- Calibrating VB-based early-stopping rules against MCMC benchmarks to support computationally cheap interim analyses in sequential clinical-trial designs.
- Packaging the algorithm as an open-source R package for reproducible trial simulation.

Technical Skills

Languages	R (advanced), Python (advanced), SQL, Bash/Shell, SAS [edit or delete]
Bioinformatics	Next-generation sequencing pipelines, variant calling (`iVar`, `Medaka`, `BCFtools`), consensus-sequence inference, genomic data QC
Software & Tools	R packages (`tidyverse`, `Stan`/`rstan`, `survival`, `devtools`), Python (`NumPy`, `pandas`, `PyTorch`, `scikit-learn`), Git/GitHub, HPC/SLURM clusters, LaTeX/Beamer, R Markdown & Quarto, Docker [edit or delete]
Practices	Reproducible research, simulation study design, statistical software development, technical writing & scientific communication

Presentations

July 2026 *A Novel Profile Hidden Markov Model for Genotyping Viruses in Diagnostic Samples from Noisy Sequencing Reads.* Research group seminar, Department of Statistics, Iowa State University.

Certifications

Business Analytics with Excel
Foundations: Data Everywhere
Introduction to Clinical Research
Introduction to Informed Consent
ICH Good Clinical Practice
Essential Elements of Ethics in Clinical Research
Introduction to Data Management for Clinical Research
Introduction to Statistics and Data Analysis in Public Health
Measuring disease in Epidemiology
Python for Genomic Data Science; Introduction to Genomics Technologies

REFERENCES

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